

COMPUTER NETWORKS

ASSIGNMENT

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Q1. Develop Static Routing with 4 PC's, 4 Switches & 8 PC's. Make sure to highlight the Networks with their Network ID's. IP Addresses should be Written in text format as well. (6 M)

Answer:

Static routing is a method in which routes to different networks are manually configured on routers by the network administrator. In the given Cisco Packet Tracer topology, four routers, four switches and eight PCs are interconnected to form four LAN networks.

Each LAN contains two PCs connected to a switch, while the switch is connected to its respective router. The four routers are connected sequentially through three point-to-point links. Static routes are then configured so that packets can travel from one LAN to another through the appropriate routers.

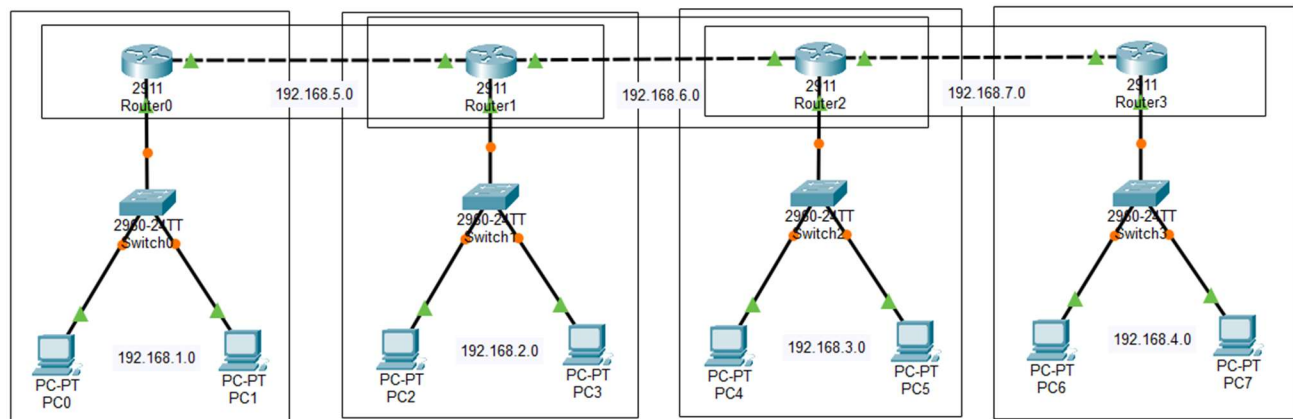


Figure 1: Static routing network topology

1. Network Identification

The topology is divided into four LAN networks and three router-to-router networks. The Network IDs shown in the topology are listed below.

Network	Network ID	Connected Devices / Purpose
LAN 1	192.168.1.0	Router 0, Switch 0, PC0 and PC1
LAN 2	192.168.2.0	Router 1, Switch 1, PC2 and PC3
LAN 3	192.168.3.0	Router 2, Switch 2, PC4 and PC5
LAN 4	192.168.4.0	Router 3, Switch 3, PC6 and PC7
Link 1	192.168.5.0	Connection between Router 0 and Router 1
Link 2	192.168.6.0	Connection between Router 1 and Router 2
Link 3	192.168.7.0	Connection between Router 2 and Router 3

2. IP Addressing

The topology identifies the Network IDs but does not specify the individual host/interface addresses in the submitted assignment. The following addressing plan is therefore used as a consistent example for the same topology. A /24 subnet mask is used for the LANs and a /30 subnet mask is used for the point-to-point router links.

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.3	255.255.255.0	192.168.1.1
Router 0	LAN	192.168.1.1	255.255.255.0	—
Router 0	R0-R1	192.168.5.1	255.255.255.252	—
Router 1	LAN	192.168.2.1	255.255.255.0	—
Router 1	R0-R1	192.168.5.2	255.255.255.252	—
Router 1	R1-R2	192.168.6.1	255.255.255.252	—
PC2	NIC	192.168.2.2	255.255.255.0	192.168.2.1
PC3	NIC	192.168.2.3	255.255.255.0	192.168.2.1
Router 2	LAN	192.168.3.1	255.255.255.0	—
Router 2	R1-R2	192.168.6.2	255.255.255.252	—
Router 2	R2-R3	192.168.7.1	255.255.255.252	—
PC4	NIC	192.168.3.2	255.255.255.0	192.168.3.1
PC5	NIC	192.168.3.3	255.255.255.0	192.168.3.1
Router 3	LAN	192.168.4.1	255.255.255.0	—
Router 3	R2-R3	192.168.7.2	255.255.255.252	—
PC6	NIC	192.168.4.2	255.255.255.0	192.168.4.1
PC7	NIC	192.168.4.3	255.255.255.0	192.168.4.1

3. Static Route Configuration

After assigning the IP addresses, static routes are configured on each router. A static route specifies the destination network, subnet mask and next-hop IP address. This enables the router to forward packets to networks that are not directly connected.

Router	Static Route	Purpose
R0	ip route 192.168.2.0 255.255.255.0 192.168.5.2	Route to LAN 2 through R1
R0	ip route 192.168.3.0 255.255.255.0 192.168.5.2	Route to LAN 3 through R1

R0	ip route 192.168.4.0 255.255.255.0 192.168.5.2	Route to LAN 4 through R1
R1	ip route 192.168.1.0 255.255.255.0 192.168.5.1	Return route to LAN 1
R1	ip route 192.168.3.0 255.255.255.0 192.168.6.2	Route to LAN 3 through R2
R1	ip route 192.168.4.0 255.255.255.0 192.168.6.2	Route to LAN 4 through R2
R2	ip route 192.168.1.0 255.255.255.0 192.168.6.1	Route to LAN 1 through R1
R2	ip route 192.168.2.0 255.255.255.0 192.168.6.1	Route to LAN 2 through R1
R2	ip route 192.168.4.0 255.255.255.0 192.168.7.2	Route to LAN 4 through R3
R3	ip route 192.168.1.0 255.255.255.0 192.168.7.1	Route to LAN 1 through R2
R3	ip route 192.168.2.0 255.255.255.0 192.168.7.1	Route to LAN 2 through R2
R3	ip route 192.168.3.0 255.255.255.0 192.168.7.1	Route to LAN 3 through R2

4. Working of Static Routing

When a PC sends a packet to a device in another LAN, the packet is first sent to the default gateway of the source network. The router checks its routing table and forwards the packet to the next-hop router. The packet continues through the intermediate routers until it reaches the router connected to the destination LAN. The destination router then forwards the packet to the appropriate PC through its switch.

5. Verification

After configuring the routers and PCs, connectivity can be tested using the ping command. For example, a ping can be sent from PC0 (192.168.1.2) to PC7 (192.168.4.3). Successful replies indicate that the IP addresses, default gateways, router interfaces and static routes have been configured correctly.

Conclusion

Thus, static routing establishes communication between the four different LANs by manually defining the paths through which packets must travel. The routers use their routing tables to forward packets from the source network toward the appropriate destination network through the intermediate routers. The topology therefore demonstrates communication between four LANs, four switches, eight PCs and four routers using static routing.

Q2. What is OSI model? What are the different layers of OSI model? Write the functionality, Data format and Protocols used at each layer. (3 M)

Answer:

The OSI (Open Systems Interconnection) model is a seven-layer reference model used to describe the process of communication between computers over a network. It was developed to provide a common framework for understanding network communication and to make different networking technologies easier to design, implement and troubleshoot. Each layer has a specific responsibility and provides services to the layer above it.

The seven layers of the OSI model are Application, Presentation, Session, Transport, Network, Data Link and Physical. The layers work together, but each layer performs a different task. During transmission, information moves from the Application layer toward the Physical layer. At the receiving system, the process occurs in the reverse direction.

Layers of the OSI Model

Layer	Functionality	Data Format	Protocols / Examples
7. Application	Provides network services used by applications such as web browsing, email, file transfer, remote login and name resolution.	Data / Message	HTTP, HTTPS, FTP, SMTP, DNS, DHCP, SSH, Telnet
6. Presentation	Translates data into a suitable format and handles encryption, decryption, compression and decompression.	Data	TLS/SSL, JPEG, PNG, MPEG, ASCII, Unicode
5. Session	Establishes, manages and terminates sessions between applications. It also provides dialog control and synchronization.	Data	RPC, NetBIOS and session-management mechanisms
4. Transport	Provides end-to-end delivery. Performs segmentation, reassembly, flow control, reliability and port-based process identification.	Segment / Datagram	TCP, UDP
3. Network	Provides logical addressing and determines routes for packets between different networks.	Packet	IP, ICMP, IPsec, OSPF, RIP, BGP
2. Data Link	Provides framing, MAC addressing, local delivery, media access control and error detection.	Frame	Ethernet, IEEE 802.11 Wi-Fi, PPP, HDLC
1. Physical	Transmits raw bits over the physical medium and defines signaling, cables,	Bits	Ethernet physical standards, fiber, copper, USB, DSL

	connectors and physical transmission characteristics.		
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Explanation of the Seven Layers

- 1. Application Layer:** The Application layer is the top layer of the OSI model and is closest to the user. It provides services that network applications need in order to communicate. For example, a web browser uses HTTP or HTTPS to request information from a web server. Email applications can use SMTP for sending mail, while DNS is used to translate domain names into IP addresses.
- 2. Presentation Layer:** The Presentation layer is concerned with the representation of data. Different systems may store or represent information in different formats, so this layer helps make the data understandable to both sides. It can perform data translation, encryption and decryption, and compression and decompression. Examples include character encoding and data formats such as JPEG and PNG.
- 3. Session Layer:** The Session layer manages communication sessions between two applications. It establishes a session, maintains it while communication is taking place and terminates it after communication is complete. It can also provide synchronization points so that communication can be organized into separate parts.
- 4. Transport Layer:** The Transport layer provides end-to-end communication between applications running on different systems. It breaks large messages into smaller units and reassembles them at the destination. TCP provides reliable communication using sequencing, acknowledgements and retransmission, while UDP provides faster connectionless communication with less overhead. Port numbers are used to identify the destination application.
- 5. Network Layer:** The Network layer is responsible for logical addressing and routing. IP addresses are used to identify source and destination systems, while routers use routing information to determine where packets should be forwarded. Routing protocols such as OSPF, RIP and BGP help exchange or calculate routing information.
- 6. Data Link Layer:** The Data Link layer is responsible for communication between directly connected devices. It organizes data into frames, uses MAC addresses for local delivery, controls access to the transmission medium and can perform error detection. Ethernet and Wi-Fi both use Data Link layer mechanisms.
- 7. Physical Layer:** The Physical layer deals with the actual transmission of bits through a medium. It specifies physical characteristics such as electrical or optical signals, connectors, cables, frequencies and transmission rates. Examples include copper Ethernet cables and fiber-optic communication.

Encapsulation and Decapsulation

When a sender transmits data, the information passes downward through the OSI layers. The Transport layer divides the data into segments or datagrams. The Network layer adds logical addressing and forms packets. The Data Link layer adds framing information and forms frames. Finally, the Physical layer converts the information into bits for transmission.

At the receiving end, the reverse process takes place. The Physical layer receives the bits, the Data Link layer processes the frame, the Network layer processes the packet, and the Transport layer reassembles the data. The upper layers then deliver the information to the correct application. This reverse process is known as decapsulation.

The OSI model is also useful in troubleshooting. For example, a damaged cable is mainly associated with the Physical layer, an incorrect MAC or frame-related problem may involve the Data Link layer, an incorrect IP address or routing table can affect the Network layer, and a blocked port can involve the Transport layer.

Q3. What are the different Error Detection methods? Explain each with the help of proper examples. (3 M)

Answer:

During data transmission, the information travelling through a communication channel may be affected by noise, interference, attenuation or other disturbances. Because of these factors, one or more bits can change from 0 to 1 or from 1 to 0. Error detection techniques add extra information to the transmitted data so that the receiver can check whether the received data has been corrupted.

The important error detection methods are Parity Check, Checksum, Cyclic Redundancy Check (CRC) and Longitudinal Redundancy Check (LRC). These methods mainly detect errors; they do not necessarily correct the corrupted data.

1. Parity Check

Parity checking is one of the simplest error detection methods. An additional bit called a parity bit is added to the original data. In even parity, the parity bit is selected so that the total number of 1s becomes even. In odd parity, the parity bit is selected so that the total number of 1s becomes odd.

Example: Consider the data 1011001. There are four 1s in this data. Since four is already even, the even parity bit is 0. Therefore, the transmitted data becomes 10110010. If one bit changes during transmission, the number of 1s may become odd, allowing the receiver to detect the error.

- Advantages: simple, fast and requires very little additional information.
- Limitation: it may not detect an even number of bit errors because the parity can remain unchanged.

2. Checksum

Checksum is an arithmetic-based error detection technique. The sender divides the data into fixed-size blocks or words and adds them using one's-complement arithmetic. The complement of the final sum is taken as the checksum and is sent along with the original data.

At the receiver, the received data words and checksum are added again. If the result satisfies the expected condition, the data is considered valid. If the result is different, the receiver detects an error.

Example: Consider two 4-bit data words, 1010 and 1100. Their binary sum is 1 0110. The carry is added back to the lower four bits: $0110 + 1 = 0111$. The one's complement of 0111 is 1000. Therefore, 1000 can be used as the checksum. The receiver performs the same type of calculation to check the received information.

- Advantages: simple arithmetic implementation and useful for detecting many transmission errors.
- Limitation: some different error patterns can produce the same checksum.

3. Cyclic Redundancy Check (CRC)

CRC is a widely used and comparatively powerful error detection method. In CRC, the binary data is treated as a polynomial and divided by a fixed generator polynomial using modulo-2 division. The remainder obtained from the division is appended to the original data before transmission.

At the receiver, the complete received codeword is divided by the same generator. If the remainder is zero, the data is assumed to be valid. If a non-zero remainder is obtained, an error is detected.

Example: Let the data be 1101 and the generator be 1011. Since the generator contains four bits, three zeros are appended to the data, giving 1101000. This value is divided using modulo-2 division by 1011. The resulting remainder is appended to 1101 to form the transmitted codeword. The receiver repeats the division using the same generator and checks the remainder.

- CRC is especially effective for detecting burst errors.
- CRC is commonly used in networking frames and many digital communication systems.
- The generator polynomial determines the error patterns that can be detected.

4. Longitudinal Redundancy Check (LRC)

LRC is an extension of the parity concept in which data is arranged in a block of rows and columns. Instead of generating only one parity bit for one data unit, parity is calculated across corresponding bit positions. The resulting parity values form an additional row called the LRC.

Example: Suppose several binary bytes are placed one below another. The parity of each column is calculated and the resulting parity bits are placed in an additional row. At the receiver, the column parity is calculated again. If the calculated parity does not agree with the transmitted LRC, the receiver knows that an error has occurred.

- Advantages: can detect many errors across a block of data and provides stronger checking than a single parity bit.
- Limitation: certain combinations of errors can still go undetected.

Comparison of Error Detection Methods

Method	Basic Principle	Example / Use	Limitation
Parity Check	Adds one parity bit to maintain even or odd parity.	1011001 with even parity uses parity bit 0.	May fail to detect an even number of bit errors.
Checksum	Uses one's-complement arithmetic to generate a check value.	1010 + 1100 produces a checksum of 1000.	Some different error patterns may have the same checksum.
CRC	Uses modulo-2 polynomial division and transmits the remainder.	1101 processed using generator 1011.	Requires polynomial-based calculation.
LRC	Calculates parity across columns of a block.	Parity values are calculated for each column of multiple bytes.	Some complex error patterns may remain undetected.

Conclusion

Error detection is an important part of reliable computer communication because transmitted data can be changed by noise and interference. Parity Check provides a simple method for detecting basic errors, Checksum uses arithmetic operations to verify data, CRC provides strong detection using polynomial division, and LRC extends parity checking across a block of data. The choice of method depends on the type of communication system and the level of error detection required.